

False Coherence: A Thermodynamic Failure Regime in Chain-of-Thought Reasoning

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Abstract

We study the internal stability of chain-of-thought reasoning chains using a scalar metric Φ computed step by step from text embeddings alone, without access to model weights, activations, or token probabilities. Applying this metric to real model-generated reasoning chains from a step-labeled dataset, we identify two structurally distinct failure regimes. In *destabilized drift*, Φ drops at the first labeled error step, indicating that the chain degrades as it enters an incorrect state. In *false coherence*, Φ remains flat or rises at the error step despite the chain becoming incorrect, with the reasoning staying locally coherent and on-topic while producing a wrong answer. In our sample of 10 matched pairs from PRM800K, false coherence was the more common observed regime, appearing in 7 of 10 cases. We show that this finding helps explain why three variants of Φ -based self-consistency chain selection all underperformed standard majority voting in the same sample: stability-based weighting amplifies false-coherent wrong chains rather than suppressing them. We further show that Φ is operational across mathematical, factual, and quantitative reasoning domains, and produces directional evidence in retrieval-augmented generation settings. We report a null result for agentic trajectory separation in a within-instance matched cohort. All experiments use real published datasets with no synthetic data.

1 Introduction

Chain-of-thought prompting asks a language model to write out its reasoning before giving a final answer (Wei et al., 2022). The idea is that making reasoning visible improves both answer quality and interpretability. This has worked well in practice, and stepwise reasoning has become a standard technique across many tasks.

What is less well understood is what happens inside a reasoning chain when things go wrong. Existing evaluation approaches are mostly outcome-based: a chain is judged by whether its final answer is correct, by majority vote across many sampled chains (Wang et al., 2023), or by a separately trained verifier or process reward model (Lightman et al., 2023). These approaches generally do not provide a simple stepwise signal, computed from observable text alone, that tells you whether the reasoning trajectory itself is stable or degrading as it unfolds.

We built such a signal. We define a scalar stability metric Φ computed at each reasoning step from the step’s text embeddings, requiring no access to model weights, activations, attention patterns, or token probabilities. The metric combines three quantities: how well the current step stays aligned with the original question, how locally continuous the step is with the steps before it, and how much disorder has accumulated across recent steps. We describe the full formulation in Section 2.

When we applied this instrument to real model-generated reasoning chains from a step-labeled dataset (Lightman et al., 2023), we found a pattern we did not anticipate. In 7 out of 10 matched pairs we examined, the chain that reached an incorrect conclusion showed a rising or flat Φ value at the first labeled error step, not a falling one. The reasoning stayed locally coherent and on-topic while producing a wrong answer. We call this **false coherence**.

The contrasting pattern, where Φ drops at the error step and the chain visibly destabilizes, we call **destabilized drift**. In our PRM800K sample, false coherence was the more common of the two regimes, appearing in 7 of 10 evaluated pairs.

This finding has a practical consequence. Methods that use stability-like signals to weight or filter reasoning chains can amplify false-coherent wrong chains rather than suppress them. We show this empirically: three variants of Φ -based self-consistency selection all underperformed standard majority voting on our evaluation set, and the false coherence rate in the dataset explains why.

This paper makes the following contributions:

- We introduce a black-box stepwise stability metric for chain-of-thought reasoning that requires only observable text.
- We identify and characterize two thermodynamically distinct reasoning failure regimes: false coherence and destabilized drift.
- We show that false coherence was the more common observed failure mode in our PRM800K sample and that it helps explain negative results in stability-based chain selection.
- We show that Φ is operational across multiple reasoning domains and in retrieval-augmented generation, and report an exploratory null result for agentic trajectory separation.

2 The Φ Framework

2.1 Motivation

At each step, a reasoning chain can be evaluated along three structural dimensions: how well it stays connected to the original question, how locally continuous it is with the steps before it, and how focused or scattered the recent steps are. Lower values on one or more of these dimensions indicate structural degradation, even if the chain has not yet produced a wrong answer.

We want to measure these properties at each step using only the output text, so the measurement works on any model without special access to its internals.

2.2 The Metric

For each reasoning step t we compute three quantities from text embeddings of the prompt and steps:

- I_t (**prompt-alignment**): cosine similarity between step t and the original prompt, clamped to $[0, 1]$. Higher means the step is still engaging with the original question.
- ρ_t (**coherence**): cosine similarity between step t and the previous step, clamped to $[0, 1]$. Higher means the step follows naturally from what came before. For $t = 0$ we set $\rho_0 = 1.0$.
- S_t (**disorder**): mean pairwise cosine distance among the most recent up to $w = 5$ step embeddings, normalized to $[0, 1]$. Higher means the recent steps are scattered rather than focused.

The stability metric at step t is:

$$\Phi_t = I_t \times \rho_t - \alpha \times S_t \quad (1)$$

where $\alpha = 0.1$ is a fixed scaling constant. We use `sentence-transformers/all-MiniLM-L6-v2` for all embeddings throughout this work. All three quantities are computed from observable text representations only. No access to model weights, hidden states, logits, or token probabilities is required.

2.3 Step Classification

Each step is classified into one of three states based on a reference threshold $\Phi_c = 0.25$:

- **Stable:** $\Phi_t \geq \Phi_c$
- **Warning:** $0 < \Phi_t < \Phi_c$
- **Collapse:** $\Phi_t < 0$

Chain-level summary outputs include mean Φ , minimum Φ , warning-step count, collapse-step count, and the index of the first warning or collapse step.

2.4 What This Measures and What It Does Not

Φ measures structural stability: whether the reasoning stays aligned, coherent, and focused. It does not measure correctness. A chain can score high on Φ while being wrong, and can score low on Φ while being correct. This distinction is central to the findings in this paper.

3 Stepwise Monitoring of Chain-of-Thought Stability

3.1 Correct Chains Produce Stable Φ Values

We first checked whether Φ produces sensible outputs on correct reasoning chains. Using 10 GSM8K test examples with published gold rationales (Cobbe et al., 2021) and fixed dataset indices, we computed Φ at every step using the formula above with no parameter tuning.

All 10 chains remained in the stable band throughout. Mean Φ across the 10 chains was 0.520, mean minimum Φ was 0.388, and there were zero warning or collapse steps across all 36 steps evaluated.

In this GSM8K sample, the Φ signal behaved as expected for clean mathematical reasoning: it stayed comfortably above the threshold throughout each chain. These results provide a baseline for comparison with the error-transition behavior we examine next.

3.2 Φ Is Sensitive to Reasoning Errors

We then applied the metric to 10 matched correct/incorrect chain pairs from the PRM800K train split (Lightman et al., 2023), a dataset of model-generated reasoning chains with human-annotated step-level correctness labels. For each pair, we identified the first step labeled incorrect and treated that point as the error transition. We then compared mean Φ in the steps before the error transition to mean Φ at and after it.

In 7 of 10 matched pairs, mean Φ was lower in the segment at and after the error transition than in the segment before it, with a mean drop of 0.044. In this sample, Φ was therefore sensitive to reasoning degradation at the broader segment level.

However, this before-versus-after segment comparison does not capture the immediate stepwise behavior at the error transition itself. That difference becomes clear only when we look at the immediate step-level change, which we examine in Section 4.

4 False Coherence and Destabilized Drift

4.1 Two Ways a Reasoning Chain Can Fail

The segment-level analysis in Section 3 treated all failures similarly: a chain goes wrong and Φ drops on average afterward. But when we examine the immediate step-level change in Φ at the error transition, the change from the step just before the error to the step at the error, a different picture emerges.

We define the immediate step-level change as:

$$\Delta\Phi = \Phi(\text{previous step}) - \Phi(\text{error step}) \quad (2)$$

When $\Delta\Phi > 0$, Φ dropped at the error step. When $\Delta\Phi \leq 0$, Φ stayed flat or rose at the error step despite the chain becoming incorrect according to the step label.

This yields a two-regime classification:

- **Destabilized drift** ($\Delta\Phi > 0$): Φ drops at the error step. The chain degrades as it enters an incorrect state.
- **False coherence** ($\Delta\Phi \leq 0$): Φ stays flat or rises at the error step despite the chain becoming incorrect. The chain appears structurally stable while producing a wrong answer.

4.2 Results on PRM800K Matched Pairs

Applying this classification to the same 10 matched pairs from the PRM800K train split:

Regime	Count (out of 10)	Mean $\Delta\Phi$
False coherence ($\Delta\Phi \leq 0$)	7	-0.142
Destabilized drift ($\Delta\Phi > 0$)	3	+0.283

Table 1: Regime classification of 10 PRM800K matched pairs.

The two regimes are also separated in their post-error behavior. For false-coherent chains, mean post-error Φ was 0.339, indicating that the chain continued to appear stable after going wrong. For destabilized-drift chains, mean post-error Φ was 0.189, indicating continued degradation after the error. The gap between these values was 0.150.

4.3 What False Coherence Means

False coherence describes incorrect reasoning that remains structurally stable along the dimensions measured by Φ : prompt alignment, local coherence, and low disorder. The steps stay on-topic, they follow naturally from each other, and the chain does not scatter or collapse. The chain does not exhibit instability in the dimensions Φ captures, yet it is still wrong.

This differs from the intuitive picture of reasoning failure, which often assumes a chain that visibly goes off the rails. In this PRM800K sample, false coherence was the more common observed regime.

We note that this finding is based on 10 matched pairs from one dataset. Whether these proportions generalize to other datasets, models, or reasoning formats remains an open question, which we discuss in Section 9.

5 False Coherence Amplification in Naive Φ -Based Selection

5.1 Why This Matters for Chain Selection

The false coherence finding has a direct practical consequence. Methods that select, weight, or filter reasoning chains based on their Φ scores can favor false-coherent wrong chains, because such chains score high on stability along the dimensions Φ measures. We tested three variants of Φ -based self-consistency selection to see whether this effect appears in practice.

All three experiments used the same 10 prompt groups from the PRM800K train split, each with multiple sampled reasoning chains per prompt. Raw majority voting on this set achieved 70% accuracy (7 of 10 prompts correct). We use that as the baseline throughout this section.

5.2 Experiment 1: Φ -Weighted Voting

Instead of counting each chain’s vote equally, we weighted each vote by the chain’s mean Φ . Chains with higher Φ contributed more to the final answer selection.

Φ -weighted voting achieved 40% accuracy (4 of 10 prompts correct), compared to 70% for raw majority. In all 3 cases where the two methods disagreed, Φ -weighting selected the wrong answer. In each of those cases, a wrong chain with high Φ received amplified voting weight and overrode the correct majority answer. This is consistent with the false coherence finding: wrong chains that remain thermodynamically stable can receive higher weights and dominate the vote.

5.3 Experiment 2: Hard Φ Gate

We excluded all chains with mean $\Phi < \Phi_c = 0.25$ before voting. Only chains that passed this stability threshold participated in the vote.

This reduced accuracy to 40% and caused 4 of 10 prompts to abstain entirely because all chains for those prompts fell below the threshold. PRM800K chains in this sample cluster near or below $\Phi_c = 0.25$, unlike the GSM8K baseline chains, which averaged 0.520. In this sample, the threshold that behaved sensibly on correct GSM8K chains was too aggressive as a filter on PRM800K chains.

5.4 Experiment 3: Collapse-Only Filter

We excluded only chains with at least one collapse step, leaving chains in the warning band eligible to vote. This was less aggressive than the hard gate: 2 of 10 prompts abstained, and accuracy reached 50%, still below the 70% raw majority baseline.

5.5 Summary

Each softer use of Φ caused less harm, but none of the three strategies improved over raw majority voting in this sample. In this PRM800K sample, where false coherence was the more common

Method	Accuracy	vs. Raw Majority
Raw majority vote	0.70	—
Φ -weighted vote	0.40	−0.30
Hard Φ gate (≥ 0.25)	0.40	−0.30
Collapse-only filter	0.50	−0.20

Table 2: Self-consistency selection results on 10 PRM800K prompt groups. All three Φ -based strategies underperformed raw majority voting in this sample.

observed regime, treating stability as a proxy for correctness can amplify false-coherent wrong chains.

These results do not show that Φ is uninformative. Rather, they show that stability alone is not a sufficient proxy for correctness in this PRM800K sample. We discuss what a combined approach might look like in Section 9.

6 Cross-Domain Generalization

6.1 Setup

We measured Φ on reasoning chains from three distinct domains to check whether the same formula, threshold, and scaling constant produce meaningful outputs beyond mathematical reasoning. The domains and chain types are:

- **GSM8K** (Cobbe et al., 2021): mathematical word problems. Native gold rationale lines from the published dataset.
- **HotpotQA** (Yang et al., 2018): multi-hop factual questions. Supporting-fact sentences extracted from human-annotated context, not native reasoning chains.
- **AQUA-RAT** (Ling et al., 2017): quantitative word problems. Native human-written rationales.

We used 10 examples per domain with fixed dataset indices. The same Φ formula, same $\alpha = 0.1$, and same threshold $\Phi_c = 0.25$ were applied across all three domains with no retraining or threshold adjustment. We note that the chain types are not identical across domains: GSM8K and AQUA-RAT use native rationale lines, while HotpotQA uses extracted supporting facts. This is a heterogeneous cross-domain measurement, not a controlled comparison of identical chain formats.

6.2 Results

Domain	Mean Φ	Mean Min Φ	Warning Steps	Collapse Steps
GSM8K (math)	0.520	0.409	0	0
HotpotQA (factual)	0.412	0.197	7	1
AQUA-RAT (quantitative)	0.326	0.061	15	5

Table 3: Φ measurements across three reasoning domains, 10 examples each. All three domains produce mean Φ above $\Phi_c = 0.25$.

All three domains produced mean Φ above the reference threshold of 0.25 with no parameter adjustment. This does not mean all chains or all steps were stable; HotpotQA and AQUA-RAT still showed warnings and collapse events. Stability strength varied across domains: GSM8K gold rationales were strongly stable with zero warnings, while AQUA-RAT chains showed more warnings and collapses. The noisier AQUA-RAT behavior may reflect limitations of the embedding model when handling mathematical notation and symbolic expressions, rather than a clear failure of the Φ framework itself.

6.3 What This Shows and What It Does Not

In this sample, the Φ framework produced above-threshold mean stability values across three structurally different reasoning domains using the same parameters. This suggests the metric is operational as a cross-domain measurement instrument.

We do not claim this shows domain-independent behavior in a strong sense. The sample is small (10 examples per domain), the chain types differ, and we have not tested domains outside mathematical and factual reasoning. What the results do show is that the same formula and threshold produced sensible outputs across the domains we evaluated without any domain-specific tuning.

7 Extension to RAG Stability

7.1 Setup

Retrieval-augmented generation (RAG) systems generate responses by combining a question with retrieved documents. Some sentences in the response are supported by the retrieved documents; others are not. We tested whether Φ can distinguish these two conditions at both the response level and the individual sentence level.

We used the ExpertQA subset of the ragbench dataset (Rungalileo, 2024), which provides multi-sentence responses with sentence-level annotations indicating which sentences are supported by the retrieved documents. We treated each response sentence as a monitored reasoning step, with the question as the prompt.

We evaluated 10 fully grounded responses (all sentences supported) and 10 partially ungrounded responses (at least one sentence not supported), using fixed stream indices from the test split. The same Φ formula, same $\alpha = 0.1$, and same threshold $\Phi_c = 0.25$ were applied with no adjustment.

7.2 Results

At the response level:

Group	Mean Φ	Mean Min Φ	Mean Warning Rate
Grounded responses	0.400	0.275	0.345
Ungrounded responses	0.321	0.273	0.473
Delta	+0.079	+0.001	—

Table 4: Response-level Φ for grounded vs. ungrounded RAG responses, 10 examples each.

At the sentence level, we compared the mean of per-example supported-sentence Φ values to the mean of per-example unsupported-sentence Φ values:

Sentence Type	Mean Φ
Supported sentences	0.399
Unsupported sentences	0.343
Delta	+0.056

Table 5: Mean of per-example sentence-level Φ values at supported vs. unsupported sentences across 20 examples.

Grounded responses showed higher mean Φ than ungrounded ones (0.400 vs. 0.321, delta 0.079), and lower warning rates (0.345 vs. 0.473). At the sentence level, supported sentences showed higher mean Φ than unsupported ones (0.399 vs. 0.343, delta 0.056).

7.3 Interpretation

In this sample, Φ distinguished grounded from partially ungrounded responses at both the response level and the sentence level. The sentence-level result suggests the metric may be sensitive to grounding status at the sentence level, though this analysis is not a strict within-response matched comparison.

We note two limitations. First, the response-level comparison is across different questions. Grounded and ungrounded examples may differ in question type or response length in ways that independently affect Φ . Second, the sentence-level analysis is likewise not a within-question or within-response matched causal test. Both results are directional evidence from a small sample of 10 examples per group, not definitive findings.

8 Exploratory Agentic Trajectory Measurement

8.1 Setup

We applied Φ to real software engineering agent trajectories from the nebius/swe-agent-trajectories dataset (Yang et al., 2024), which contains sequences of agent reasoning and action steps directed at solving GitHub issues. Each trajectory was treated as a monitored step sequence, with the task description as the prompt.

We selected 10 successful trajectories and 10 failed trajectories from the same task instance (`Melevir_cognitive_complexity-15`), using fixed stream indices. Restricting to a single task instance was necessary to avoid confounding the comparison by task identity. An earlier analysis that mixed trajectories from different tasks produced an apparent signal that disappeared when we corrected for this.

8.2 Results

Within this matched within-instance cohort, we observed no meaningful difference in mean Φ between successful and failed trajectories (delta -0.003). Warning rates were also nearly identical (0.270 vs. 0.257).

8.3 Interpretation

This is a null result. Within this single task instance, Φ did not separate successful from failed agent trajectories on either mean stability or warning rate.

Group	Mean Φ	Mean Warning Rate	Mean Steps
Successful trajectories	0.337	0.270	20.9
Failed trajectories	0.340	0.257	14.6
Delta	-0.003	—	—

Table 6: Φ measurements for successful vs. failed agent trajectories, 10 examples each, within a single task instance.

We note that this result is narrow in scope: one task instance, one agent system, 20 total trajectories. It does not show that Φ is uninformative for agentic settings in general. It shows that in this particular matched cohort, the stability signal did not distinguish outcome.

The Φ framework is operational on agentic trajectories in the sense that it runs, produces valid step-level outputs, and computes well-formed chain summaries. Whether it separates successful from failed trajectories across a broader and more diverse set of tasks remains an open question. We include this result because it is honest to report: an earlier cross-task comparison produced an apparent positive signal that did not survive proper matching.

9 Discussion, Limitations, and Future Work

9.1 What This Work Shows

The central finding of this paper is that reasoning chain failures are not all the same kind of thing. In our PRM800K sample, false coherence was the more common observed failure regime, with wrong chains that remained thermodynamically stable at the moment they went wrong, staying prompt-aligned, locally coherent, and low-disorder while producing an incorrect conclusion.

This finding has a practical consequence that we demonstrated empirically: naive use of Φ as a chain selector amplifies false-coherent wrong chains rather than suppressing them. All three selection strategies we tested underperformed raw majority voting, and the false coherence rate in the dataset provides a mechanistic explanation for why.

We also showed that Φ is operational as a measurement instrument across mathematical, factual, and quantitative reasoning domains, and showed directional evidence in the RAG setting. In an agentic setting, the framework ran successfully but did not separate successful from failed trajectories within our matched cohort.

9.2 Limitations

Sample size. The false coherence finding is based on 10 matched pairs from one dataset. The 7/10 proportion is an observation, not an established rate. Whether this proportion holds in other datasets, with other models, or across other reasoning types is unknown.

Dataset scope. All experiments used published benchmark datasets. We have not tested Φ on open-ended reasoning, legal reasoning, medical reasoning, or coding tasks outside the agentic trajectory setting.

Embedding model dependency. All results depend on the `all-MiniLM-L6-v2` embedding model. Different embedding models may produce different stability profiles. The noisier AQUA-RAT results may partly reflect this model’s limitations with mathematical notation.

Chain type heterogeneity. The cross-domain experiments used different chain types across domains. GSM8K and AQUA-RAT used native rationales; HotpotQA used extracted supporting

facts. This limits how directly the domain results can be compared.

Correctness signal required for regime classification. Classifying false coherence vs. destabilized drift requires knowing where the error transition is, which requires a step-level correctness signal such as a process reward model or human annotation. Φ alone, without that signal, can only measure stability, not classify failure regime.

Agentic null result scope. The agentic null result is from one task instance with 20 total trajectories. It does not generalize beyond that cohort.

9.3 What Φ Is and Is Not

Φ is a stability signal. It measures whether a reasoning chain stays aligned, coherent, and focused. It is not a correctness oracle. A chain can be stable and wrong. A chain can be unstable and right. The false coherence finding is the clearest illustration of this distinction.

These results suggest that Φ is better interpreted as a structural health indicator than as a standalone quality score. It may be most useful as one input among several, for example combined with a separate correctness or verification signal, rather than as a replacement for outcome-based evaluation.

9.4 Future Work

The most direct extension of this work is to test whether the false coherence rate observed here holds across larger samples and different datasets. A sample of 10 pairs is too small to draw strong conclusions about prevalence.

A second direction is to explore whether Φ combined with a correctness signal, such as a process reward model score, can improve chain selection in ways that Φ alone cannot. The negative selection results in this paper suggest that stability alone is not sufficient, but stability as a gate combined with a separate correctness signal as a ranker may behave differently.

A third direction is to test Φ on agentic trajectories across a broader and more diverse set of task instances, to determine whether the null result observed here is specific to the task instance studied or reflects a more general pattern.

Patent Disclosures

The methods described in this paper are the subject of U.S. Provisional Patent Application No. 64/038,659 (filed April 14, 2026). No license to implement or commercialize the described methods is granted by this publication. All rights reserved.

Code Availability

The reference implementation, validation test scripts, and audit artifacts are available in the public repository at <https://github.com/Wise314/cot-phi-stability>.

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